

CSE 491 — Project Proposal

Project title	NutriScan: A Verifiable, Constraint-Aware Household Food Planning Assistant
Group no.	[GROUP NO]
Project supervisor	Res. Asst. Dr. Taha Yiğit Alkan
Submission date	[DATE]

Group members

#	Name and surname	Student no.	Role / responsibility in the project
1	Levent Can Ceylan	[NO]	Architecture and project infrastructure; Household Planning Engine (optimisation); assistant orchestration; Privacy Gateway; decision record and observability; experiments E1–E2
2	Şükran Hilal Hocaoglu	[NO]	Safety rule engine (allergens, health rules); allergen data and gold set; household, identity and consent; catalogue and prices; pantry; recommendation; admin backend
3	Ozan Karadaş	[NO]	Mobile, web and admin clients; recipe content and health-rule sources; label-reading demo

1. Summary

Households in Türkiye make the same tiring decision every week: what to cook, what to buy, where to buy it, what is already at home and what is suitable for whom. They do it under high food inflation (33.79% year-on-year, August 2026) and with different hard constraints inside the same home, such as a child's nut allergy, a parent's coeliac disease or a grandparent's diabetes. Existing tools solve one dimension at a time: product scores (Yuka, local label scanners), prices (marketfiyati.org.tr) or a single retailer's basket (Migros MAYA AI). NutriScan is a chain-independent assistant that plans a household's week with a safety-first order: hard constraints first, then health-condition rules and goals, then household preferences and pantry, and cost last, with the budget as a limit rather than the goal. Its core is the Household Planning Engine, a single integer optimisation model that jointly selects weekday dinners, the shopping list and at most two chains while using pantry items before expiry, respecting the budget and never violating any member's hard constraint. A deterministic rule engine decides product suitability per member for allergens and for self-declared health conditions (diabetes, hypertension, coeliac disease, pregnancy) using source-cited, versioned nutrient and ingredient rules, with four honest outcomes including "could not verify". Retrieval-augmented generation (RAG) only proposes ingredient mappings and quotes rule sources; an LLM-based assistant only interprets requests and explains results, and every decision is traceable. This semester we deliver a working mobile, web and admin prototype with the verified core, the planning engine on real data and first experiment results.

2. Problem Definition and Motivation

Problem. Planning a week of meals and groceries for a household is a combinatorial decision with many interacting constraints: each member's allergens, health conditions such as diabetes or hypertension, and diet, nutrition goals (e.g., less sugar or salt), a tight budget, items at home that are about to expire, package sizes, and different prices in different chains. People solve it by habit and by reading labels one product at a time, which does not scale to a whole week or a whole household.

Why it matters. Food inflation is the largest contributor to consumer inflation in Türkiye (33.79% in August 2026 [1]); a healthy diet for a family of four costs about 133% of the net minimum wage [2]; average salt intake is about twice the national recommendation [3]; and allergen precautionary labelling ("may contain") is voluntary, so its absence does not mean a product is suitable [4]. Mistakes are costly: an unsuitable product for an allergic child is a safety issue, not an inconvenience.

Gap. Consumer apps (Yuka, Fig, Turkish label scanners) score a single product; price comparators ignore health; retailer assistants such as Migros MAYA AI plan meals but only for their own chain and treat dietary needs as filters [5]. Discount chains, which have far more stores, have no such assistant. No tool plans the week for the whole household across chains with verifiable decisions.

Target users. Households with at least one hard dietary constraint (allergy, coeliac disease) or a health condition that calls for nutrient limits (diabetes, hypertension, pregnancy), households shopping at discount chains and local markets, and households buying from more than one chain.

3. Objectives, Scope and Success Criteria

Objectives

1. Build a deterministic, explainable suitability engine for Turkish food products that covers allergens and health-condition rules (source-cited nutrient thresholds and ingredient rules), with four outcomes and a full decision record.
2. Formulate and implement the Household Planning Engine: a joint menu–basket–market integer model (weekday dinners, pack-size integrality, pantry and expiry, ≤ 2 chains, limited number of swaps) solved with an exact solver under a time limit.
3. Build a verifiable assistant in which an LLM orchestrates tools but can never change a decision, a RAG component that suggests ingredient mappings and cites rule sources without deciding, and a privacy layer that keeps household-context LLM processing in Türkiye (application data hosted in the EU, minimised and encrypted).
4. Create a clean-licensed dataset of Turkish home recipes with an allergen ontology, a verified product catalogue for two chains, and an allergen gold test set.
5. Deliver mobile, web and admin clients and evaluate the system in a beta with real households.

Out of scope

Calorie estimation from plate photos, continuous glucose monitoring, placing orders in retailers' carts, a full dietitian portal, medical advice or diagnosis, kidney-disease and drug–food interaction rules, and an Android store release in this academic year.

Success criteria

- Safety gate: zero false negatives on the allergen gold set ($n \geq 300$), 100% agreement of health-condition decisions with the threshold table on their test set, and no hard-constraint violation in any generated plan or swap (property tests over $\geq 10,000$ random households).
- Planning (E1): the joint model is compared with sequential "menu first, then list" baselines and with a metaheuristic (NSGA-II [15]) over 20 scenarios $\times$ 30 seeds; we report optimality gap, hypervolume and runtime.
- Assistant (E2): compared with an LLM-only planner on 50–100 household scenarios; we report constraint violations, budget violations, invented prices, tool-call accuracy and prompt-injection robustness; RAG ingredient mapping is compared with exact/fuzzy matching on the gold set (precision, recall).
- Beta (E3, spring): 20–40 households (≥ 10 with a hard constraint); plan/swap acceptance, planned vs. paid amount, use of expiring pantry items, week-4 retention.
- Performance: shelf decision $p95 \leq 1.5$ s; swap suggestion $p95 \leq 1$ s.

4. Similar Systems and Background

System / work	What it does	Limitation or gap your project addresses
Yuka, Fig, Ürün Dedektörü [6][7][8]	Scan a product; score it or warn for personal diet/allergen filters	Single product; no multi-member weekly plan, budget or pantry
Migros MAYA AI / meal planner [5]	Conversational meal planning and cart filling for Migros	Single chain; dietary needs as filters; no member-level verification or pantry; no cross-chain optimisation
marketfiyati.org.tr [9]	Price comparison across chains	No health, allergen or household model
Diet optimisation (Stigler; Maillot et al.) [10][11]	LP models of cheapest adequate diet; minimal departure from observed diet	Population or individual diets, not a real household basket with pack sizes, pantry and chains

Meal planning with pantry (van Rooijen et al.) [12]	Menu and purchase optimisation to reduce waste	Single store; no allergen verification or assistant
LLM benchmarks (NutriBench, FoodGuardBench) [13] [14]	LLMs unreliable for food-safety decisions	Motivates our design: engines decide, LLM explains

5. Preliminary Requirements

Functional requirements

Baseline below; the full list (29 items) is kept in the repository.

ID	Requirement	Priority (must / should / could)
FR-1	The system shall let a user create a household, invite adult members who give their own consent, and add child profiles under guardian consent.	Must
FR-2	The system shall store, per member, hard constraints (14 regulated allergens, coeliac/gluten, diet choices), optional self-declared health conditions (diabetes, hypertension, pregnancy) and soft goals separately; it shall never diagnose.	Must
FR-3	The system shall decide product suitability per member with a deterministic rule engine and return one of four outcomes: Not suitable, Caution, No conflict found, Could not verify; it shall never label a product as "safe".	Must
FR-4	The system shall apply versioned, source-cited health-condition rules (nutrient thresholds per 100 g, ingredient rules), report "Caution" with the nutrient fact and source, and use informational wording only.	Must
FR-5	The system shall scan EAN/UPC barcodes on mobile and explain each decision (matched ingredient, rule and version, data source and date, confidence).	Must
FR-6	The system shall plan weekday dinners, the shopping list and the chain choice in a single optimisation model in safety-first order (hard constraints, health rules and goals, preferences and pantry, then cost within budget) and report the optimality gap.	Must
FR-7	The system shall propose at most k swaps ($k = 1, 3, 5$) for a list and, when a plan is infeasible, suggest which soft limits could be relaxed (never hard constraints).	Must
FR-8	The system shall provide a text assistant that answers common intents by calling the decision engines, shows its steps and never changes a decision; medical/dose questions get a fixed safety response.	Must
FR-9	The system shall record every decision in an append-only decision record, keep a tamper-evident audit log, and let moderators approve catalogue corrections with a reason.	Must
FR-10	The system shall maintain a verified catalogue (Migros, then A101) with pack sizes, nutrition tables, prices and price age, and track the pantry (barcode, e-invoice/receipt, "finished").	Must
FR-11	The system shall suggest mappings from unseen label ingredient names to the dictionary with semantic search (RAG), apply them only after moderator approval, and quote rule sources in explanations.	Should
FR-12	The system shall generate a proactive weekly plan that changes nothing until approved, split the list across at most two chains, and accept natural-language constraints that are read back for confirmation.	Should
FR-13	The web app shall provide a Planning Studio with cost–health trade-offs and a "why is the plan like this" view.	Should

Non-functional requirements

ID	Requirement	How it will be checked
NFR-1	Zero allergen false negatives on the gold set; health rules agree with the threshold table; no hard-constraint violation.	CI regression on the gold sets; property-based tests
NFR-2	Shelf decision $p95 \leq 1.5$ s; swap $p95 \leq 1$ s; re-plan ≤ 3 s with gap badge.	Load tests; OpenTelemetry metrics
NFR-3	No health data in logs/traces/crash reports; household-context LLM calls processed in Türkiye; minimised and encrypted health fields.	Nightly canary test; egress review
NFR-4	100% of user-visible decisions have a decision record; the assistant cannot alter a decision.	UI-record contract test; LLM eval harness in CI
NFR-5	Module boundaries and architecture rules enforced automatically.	Spring Modulith verify + ArchUnit in CI
NFR-6	OAuth2/PKCE, household-level isolation, admin MFA, no secrets in the repository.	IDOR tests, gitleaks, dependency scanning

6. Method and Technology Plan

Approach. Research-first, contract-first development in two-week sprints. The system is a modular monolith (household, safety, catalogue, planning, recommendation, pantry, assistant, privacy, audit) whose module boundaries and architecture rules (e.g., "the LLM cannot change a decision") are verified in CI; every module has one owner.

Architecture outline. Mobile (Expo) and web/admin (React) clients use a TypeScript client generated from the OpenAPI contract → Spring Boot modular monolith → PostgreSQL (with pgvector). Inside: the safety rule engine (allergens and health-condition rules) filters and scores candidates before the Household Planning Engine (OR-Tools CP-SAT [16]) solves the joint model; the assistant (Spring AI) calls the engines as tools through a privacy gateway that routes household-context requests to EVREN (open models hosted in Türkiye); a RAG component over pgvector suggests ingredient mappings for moderators and retrieves rule sources for explanations, but is never on the decision path; every decision is written to the decision record and shown in the admin trace view.

Data. Open Food Facts (ODbL) [17] as a mirrored source; a verified catalogue for Migros and A101 (prices from marketfiyati.org.tr if permission is granted, otherwise team-collected prices with a price-age label); 200 team-written Turkish home recipes with an allergen ontology based on the Turkish Food Codex labelling regulation; a versioned health-rule threshold table with a source for every row (Turkish Food Codex nutrition-claim limits, WHO sugar and sodium guidelines [19] [20]), reviewed by a dietitian; synthetic households for development.

Layer / component	Technology or tool	Reason for the choice
Backend	Java 25, Spring Boot 4.1, Spring Modulith, Gradle	Team's common language; Boot 4.1 is the only line supported through June 2027; Modulith verifies module boundaries
Planning engine	Google OR-Tools CP-SAT (Java); Python only for experiments	Exact solver with optimality gap under a time limit (needed for E1); no extra service
Database	PostgreSQL 18 + pgvector	Relational core, JSONB decision records, vector search for RAG ingredient mapping and recipe search [21]
AI layer	Spring AI 2.0; EVREN (Türkiye) for household context; low-cost cloud models only for non-personal tasks	Data sovereignty and cost; decisions stay in deterministic engines
Mobile	Expo (React Native), expo-camera, EAS	Most mature path for barcode, speech, push and App Store release
Web + admin	Vite + React, TanStack, shadcn/ui, Recharts	Shared TypeScript and generated API client with mobile
Identity	Keycloak (self-hosted), OAuth2 PKCE	Standard, self-hostable, role-based admin
Hosting and ops	Contabo VPS, Docker Compose, Caddy, OpenTelemetry + Grafana	Low cost; reproducible deployment; observability from day one

Process	GitHub (private), Azure Boards (Scrum), AGENTS.md rules for AI coding agents, CI gates	Component ownership and automated gates prevent conflicts between different coding agents
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7. Work Plan

Week 1 = 14 September 2026; each package has a latest date and a fallback in the project calendar.

WP	Work package and its output	Responsible member(s)	Weeks
WP1	Requirements, architecture, repository, CI gates, agent rules, walking skeleton	Levent (all review)	3–7
WP2	Data factory: recipe schema, ingredient dictionary, allergen ontology, health-rule threshold table, Migros catalogue, 60 recipes with prices	Ozan (content, rule sources), Hilal (schema, catalogue)	4–11
WP3	Safety core: household and consent flows, rule engine (allergens + health conditions), gold sets, decision record integration	Hilal	7–13
WP4	Household Planning Engine v0 on real data; swaps; first E1 measurement	Levent	7–13
WP5	Mobile shelf scan with household strip and "Why?", list/swaps UI, web and admin v0	Ozan (with Hilal for admin backend)	6–14
WP6	Assistant v1 and Privacy Gateway v0; first E2 run; interim and final reports; integration	Levent (all members)	8–14

Spring semester (CSE 492): RAG ingredient mapping and source quoting, menu planner in production, pantry, proactive plan, chain split, natural-language constraints, web Planning Studio, recommendation, beta with 20–40 households, final experiments.

8. Expected Outputs and Risks

Expected outputs

By the end of this semester: working prototype (mobile + web + admin) with the verified safety core, shelf scan with explanations, list and swaps, the Household Planning Engine on real data, assistant v1; private repository with CI gates and documentation; recipe dataset (≥ 60) and allergen gold set (≥ 200); interim and final reports with first E1/E2 results.

Risks

Risk	Likelihood	Impact	Mitigation plan
Missing or wrong product/allergen data	High	High	Verified catalogue; "could not verify" instead of guessing; gold-set release gate
Price data permission not granted	Medium	Medium	Team-collected prices (~300 items, biweekly) with price age
Joint optimisation too slow at scale	Medium	Medium	Time-limited exact solver with gap badge; metaheuristic (E1)
LLM unreliable or unavailable	Medium	Medium	LLM never decides; workflow intents; template fallback
Medical-device boundary for health-condition features [22]	Medium	High	Informational wording only, no diagnosis or treatment claims, source-cited rules, dietitian review, in-app disclaimer
Health data protection (KVKK [18])	Medium	High	Minimised, encrypted health fields; none in logs; household-context LLM calls in Türkiye; EU hosting under KVKK standard contract; legal review before beta
Conflicts between AI coding agents	Medium	Medium	Component ownership, contract-first changes, CI gates

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Supervisor Approval

To be completed by the project supervisor. The proposal is submitted to MS Teams after the supervisor has reviewed it.

Supervisor name	
Assessment	Approved ☐ Approved with revisions ☐ Revise and resubmit ☐
Comments	
Date / signature	